Neighborhood Discriminative Manifold Projection for Face Recognition in Video

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***Abstract*— This paper presents a novel supervised manifold learning method called Neighborhood Discriminative Manifold Projection (NDMP) for face recognition in video. By constructing a discriminative eigenspace projection of the high-dimensional face manifold, NDMP seeks to learn an optimal low-dimensional projection by solving a constrained least-squares objective function based on local and global constraints. Local geometry is preserved through the use of intra-class and inter-class neighborhood information while global manifold structure is retained by imposing rotational invariance. The proposed method is comprehensively evaluated on a large video data set. Experimental results and comparisons with classical and state-of-art methods demonstrate the effectiveness of our method.**

***Keywords*— Manifold learning, subspace projection methods, video-based face recognition, pattern recognition.**

I. Introduction

In recent years, manifold learning has become an important topic of research in computer vision and pattern recognition. With the advancements in multimedia technology today, it plays an increasingly vital role in many applications such as face recognition in video, multimodal biometrics and human activity analysis, where data is typically represented as points in a high-dimensional space. However, it is a common presumption in manifold learning that data lies on or near a low-dimensional manifold embedded within the original high-dimensional manifold space [2]. The transformation or mapping between the high-dimensional and low-dimensional spaces can be accomplished through dimensionality reduction. Thus, dimensionality reduction is an essential task for modelling the intrinsic structure of data manifold in a more compact and meaningful embedding. In the case of face videos, the large amount of variations between face images (or frames) can be better captured through the extraction of relevant features in a low-dimensional embedded space.

Classical linear dimensionality reduction methods such Principal Component Analysis (PCA) [17], Multidimensional scaling (MDS) [5] and Linear Discriminant Analysis (LDA) [1] have been the core learning algorithms used in many computer vision and pattern recognition applications. These linear methods are clearly effective in learning data in simple Euclidean structure. PCA learns a projection that maximizes its variance while MDS preserves pairwise distances between data points in the new projected space. With additional class information, LDA learns a linear projection that maximizes the ratio of the between-class scatter to the within-class scatter.

Meanwhile, nonlinear dimensionality reduction methods such as Locally Linear Embedding (LLE) [15] and Isomap [16] began to emerge in recent years. These methods are able to discover the underlying high-dimensional nonlinear structure of the manifold in a lower-dimensional space.

The Local Linear Embedding (LLE) algorithm proposed by Saul & Roweis is an increasingly popular choice for dimensionality reduction in recent manifold learning research. Since face images can be viewed as high dimensional data points in a nonlinear manifold, the LLE algorithm seeks to learn the global structure of a nonlinear manifold through a linear reconstruction that preserves its local neighborhood structure. Isomap is similar in nature, but exploits the geodesic distances between data points in the manifold before using classic MDS [5] to reduce its dimension. The biggest drawback of these methods is that they cannot deal with the *out-of-sample* problem – whereby new data points cannot be directly projected onto the embedded space. This limits its ability to be applied to classification and recognition tasks.

In a thorough investigation, Hadid & Peitikäinen [9] showed that good recognition rates in video sequences can be produced by using LLE to build a view-based low-dimensional embedding for selection of representative faces. There are a number of supervised LLE methods in the literature, such as Supervised LLE (SLLE) by De Ridder et al. [6] and an eigendecomposition scheme for SLLE by Kouropteva et al. [12]. Although these methods elegantly retain the original weight computations from LLE, they are sensitive to perturbations due to ill-conditioned eigenvectors.

More recently, various manifold learning methods such as Locality Preserving Projections (LPP) [11], Orthogonal Locality Preserving Projections (OLPP) [3] and Neighborhood Preserving Embedding (NPE) [10] have been proposed to solve the out-of-sample problem. These methods resolve this limitation by deriving optimal linear approximations to the embedding while attempting to maintain its original local neighborhood structure using neighborhood graphs. NPE, in particular, is directly formulated based on LLE's objective functions, and therefore inherits its neighborhood preserving property. However, its lack of discriminative ability does not

effectively distinguish within-class and between-class neighborhood structures.

Other supervised methods such as Local Discriminant Embedding (LDE) [4] and Marginal Fisher Analysis (MFA) [20] further incorporated additional discriminative information to the embedding. Nevertheless, both methods utilize affinity weight matrices, which are constructed based on adjacency graphs (simple binary or heat kernel functions), not directly from actual neighborhood distances.

While many of these latest techniques reported impressive experimental results, they were evaluated on data sets with limited variations in pose, expression and illumination such as ORL, Yale and UMIST data sets [8]. Moreover, these "still-image" data sets are restricted to only a limited variety of face images. In video sequences, the abundance of face images may prove to be a challenging task due to the nonlinearity of its data manifold. A wide variety of techniques [7][9][13] have been proposed to deal with this challenge.

In this paper, we propose a novel supervised manifold learning method called Neighborhood Discriminative Manifold Projections (NDMP) for feature extraction in video-based face recognition. NDMP constructs a discriminative eigenspace projection of the face manifold based on both local intra-class and inter-class neighborhood geometry. Inspired by both neighborhood-preserving geometric properties of the LLE and discriminative properties of the LDA, our approach seeks to learn an optimal low-dimensional embedding by solving a constrained least-squares (quadratic) objective function that utilises both intra-class and inter-class neighborhood information. After some formulation, the objective function is reduced to a compact generalised eigenvalue problem, whereby a linear projection for mapping new data points to the face manifold is learned.

While the video-based face recognition framework described in this paper consists of other algorithms for data clustering and classification, the main contribution of this paper centers on the feature extraction method.

The rest of this paper is organized as follows. The formulation of the NDMP algorithm is elaborated in Section II. Section III describes the other tasks (extraction of training exemplar set, classification) that are part of the video-based face recognition process. In Section IV, we present some experimental results and comparisons with other algorithms. Finally, conclusions and suggestions for future work are provided in Section V.

II. Algorithm

In this section, we review the Locally Linear Embedding (LLE) algorithm followed by a detailed description of the proposed Neighborhood-based Discriminative Manifold Projections (NDMP) algorithm. Brief remarks on some closely-related techniques are also provided.

A. Overview of Locally Linear Embedding (LLE) Algorithm

Let $\mathbf{X} = \left\{ \mathbf{x}_i \in \Re^D \mid i = 1,..., N \right\}$ represent the input data in Euclidean space consisting of N face samples in D dimensions, belonging to c classes of faces with the corresponding class labels $\left\{ t_i \mid t_i \in \{1,...,c\}, i = 1,...N \right\}$. The Locally Linear Embedding (LLE) algorithm [15] can be summarized into 3 main steps:

Step 1: Neighborhood computation: For each point $\mathbf{x}_i$, find its k nearest neighbors.

Step 2: Computation of reconstruction weights: Compute the reconstruction weights W_{ij} that best reconstruct each point $\mathbf{x}_i$ from its neighbors by minimizing the cost function

$$\varepsilon_{rec}(\mathbf{W}) = \sum_{i=1}^{N} \left\| \mathbf{x}_i - \sum_{j=1}^{k} W_{ij} \mathbf{x}_j \right\|^2 \tag{1}$$

$\mathbf{W} = \left\{ W_{ij} \mid i = 1,...N; j = 1,...k \right\}$ is the reconstruction weight vector, subject to the constraints $\sum_{j=1}^{k} W_{ij} = 1$ and $W_{ij} = 0$ if $\mathbf{x}_i$ and $\mathbf{x}_j$ are not neighbors.

Step 3: Computation of embedding coordinates: Compute the embedding coordinates $\mathbf{y}_i$ best reconstructed by the optimal reconstruction weight vector $\mathbf{W}$ (found in Step 2), by minimizing the cost function

$$\varepsilon_{emb}(\mathbf{y}) = \sum_{i=1}^{N} \left\| \mathbf{y}_i - \sum_{j=1}^{k} W_{ij} \mathbf{y}_j \right\|^2 \tag{2}$$

constraining to $\sum_{i=1}^{N} \mathbf{y}_i = 1$ and $\sum_{i=1}^{N} \mathbf{y}_i \mathbf{y}_i^T / N = \mathbf{I}$ where $\mathbf{I}$ is an identity matrix.

Solving *arg min*(ε_{emb}) will yield the new coordinates of data points in embedded space, $\mathbf{Y} = \left\{ \mathbf{y}_i \in \Re^d \mid, i = 1,..., N \right\}$ is a $d \times N$ matrix, where the new reduced dimension of the embedded data points, $d < D$.

The primary motivation for formulating our approach based on the LLE algorithm is its neighborhood-preserving property. The global intrinsic geometry of the data points in low-dimensional embedded space is faithfully preserved through local neighborhood relations.

However, there are drawbacks to this nonlinear dimensionality reduction method. Firstly, it cannot perform direct projection of new data points to the embedded manifold, commonly known as an "out-of-sample" problem. Secondly, LLE does not possess class discriminative properties.

He et al. have thoroughly formulated a linear approximation for the projection of new data points in their Neighborhood Preserving Embedding (NPE) method [10]. Our proposed method will attempt to extend this idea farther by introducing discriminative power, while maintaining the constraint properties of the original LLE algorithm.

B. Neighborhood Discriminative Manifold Projection (NDMP)

In our proposed NDMP method, we construct two sets of reconstruction weights, unlike the NPE which has only one set

of weights. Then, the objective function is solved as a constrained minimization problem. Although generally similar to the NPE in terms of its theoretical foundation and approximation to LLE, our final objective function is different. The formulation of NDMP algorithm is elaborated as follows:

1) Construction of Intra-class and Inter-class Neighborhood Subsets:

Let all data points in a local neighborhood, $\mathbf{\Psi}$ comprise of two disjointed subsets – *intra-class* subset $\{\boldsymbol{\psi}_l \mid l = 1,\ldots,k\}$ and *inter-class* subset $\{\boldsymbol{\psi}'_m \mid m = 1,\ldots,k'\}$. Compute the k nearest *intra-class* neighbors and k' nearest *inter-class* neighbors of each point $\mathbf{x}_i$. A point $\mathbf{x}_j$ is an *intra-class* neighbor of point $\mathbf{x}_i$ if they belong to the same class $(t_i = t_j)$. Likewise, point $\mathbf{x}_j$ is an *inter-class* neighbor of point $\mathbf{x}_i$ if they do not belong to the same class $(t_i \neq t_j)$.

It is possible for a local neighborhood to have unequal number of intra-class and inter-class nearest neighbors. However, for the sake of uniformity of the weight matrices, we fixed $k = k'$. Due to class set limitation, the maximum number of intra-class and inter-class neighbors is restricted to $(M-1)$, where M is the number of data points in a class.

2) Formulation of Neighborhood Reconstruction Weights:

From the two neighborhood subsets defined earlier, their respective neighborhood reconstruction weights that best reconstruct each point $\mathbf{x}_i$ from its neighbors are computed by minimizing their respective cost functions.

Similar to Eq. (1) in Step 2 of the LLE algorithm, the intra-class reconstruction weight matrix $\mathbf{W}^r$ can be computed by minimizing the following cost function

$$\varepsilon_{rec}(\mathbf{W}^r) = \sum_{i=1}^{N} \left\| \mathbf{x}_i - \sum_{j=1}^{k} W_{ij}^r \mathbf{x}_j \right\|^2 \tag{3}$$

The inter-class reconstruction weight matrix $\mathbf{W}^e$ can be computed by minimizing the following cost function

$$\varepsilon_{rec}(\mathbf{W}^e) = \sum_{i=1}^{N} \left\| \mathbf{x}_i - \sum_{j=1}^{k} W_{ij}^e \mathbf{x}_j \right\|^2 \tag{4}$$

Both Eq. (3) and Eq. (4) can be computed in closed form by determining the optimal weights through the introduction of a local covariance matrix. Alternatively, it can also be solved using a linear system of equations.

3) Formulation of Optimal Embedding:

In this step, the optimal embedding is derived to enable new data points to be projected in the embedded space. The coordinates in the embedded space $\mathbf{Y} = \{\mathbf{y}_i \in \Re^d \mid, i = 1,\ldots,N\}$ can be computed by minimizing the cost function in Eq. (2). Similar to NPE algorithm [10], the intra-class cost function can be formulated by expanding the least-squares term:

$$\begin{aligned}
\varepsilon_{emb}^r(\mathbf{Y}) &= \sum_{i=1}^{N} \left\| \mathbf{y}_i - \sum_{j=1}^{k} W_{ij}^r \mathbf{y}_j \right\|^2 \\
&= \left\| \mathbf{Y} - \sum_{i=1}^{N}\sum_{j=1}^{k} W_{ij}^r \mathbf{y}_j \right\|^2 \\
&= tr\left\{ \left(\mathbf{Y} - \sum_{i=1}^{N}\sum_{j=1}^{k} W_{ij}^r \mathbf{y}_j \right) \left(\mathbf{Y} - \sum_{i=1}^{N}\sum_{j=1}^{k} W_{ij}^r \mathbf{y}_j \right)^T \right\} \\
&= tr\left(\mathbf{Y}[\delta_{ij} - 2\mathbf{W}^r + \left\|\mathbf{W}^r\right\|^2]\mathbf{Y}^T \right) \\
&= tr\left(\mathbf{Y}(\mathbf{I} - \mathbf{W}^r)^T(\mathbf{I} - \mathbf{W}^r)\mathbf{Y}^T \right) \\
&= tr\left(\mathbf{Y}\mathbf{M}^r\mathbf{Y}^T \right)
\end{aligned} \tag{5}$$

$\mathbf{I}$ is an identity matrix, $tr\{\cdot\}$ refers to the trace of the matrix, and the orthogonal intra-class weight matrix

$$\mathbf{M}^r = (\mathbf{I} - \mathbf{W}^r)^T(\mathbf{I} - \mathbf{W}^r) \tag{6}$$

Similarly, the inter-class cost function can be derived as

$$\varepsilon_{emb}^e(\mathbf{Y}) = tr\left(\mathbf{Y}\mathbf{M}^e\mathbf{Y}^T\right) \tag{7}$$

where the orthogonal inter-class weight matrix

$$\mathbf{M}^e = (\mathbf{I} - \mathbf{W}^e)^T(\mathbf{I} - \mathbf{W}^e) \tag{8}$$

In order to project new data, a linear projection $\mathbf{Y} = \mathbf{A}^T\mathbf{X}$ that maps the original data points $\mathbf{X}$ to the projected coordinates in embedded space $\mathbf{Y}$ using a linear transformation matrix $\mathbf{A} = \{\mathbf{a}_i \in \Re^D \mid i = 1,\ldots,d\}$ is applied to both cost functions, yielding

$$\varepsilon_{emb}^r(\mathbf{X}) = tr\left(\mathbf{A}^T\mathbf{X}\mathbf{M}^r\mathbf{X}^T\mathbf{A}\right) \tag{9}$$

$$\varepsilon_{emb}^e(\mathbf{X}) = tr\left(\mathbf{A}^T\mathbf{X}\mathbf{M}^e\mathbf{X}^T\mathbf{A}\right) \tag{10}$$

In order to model the constrained minimization problem, an objective function must be minimized subject to constraints on the possible values of independent variables.

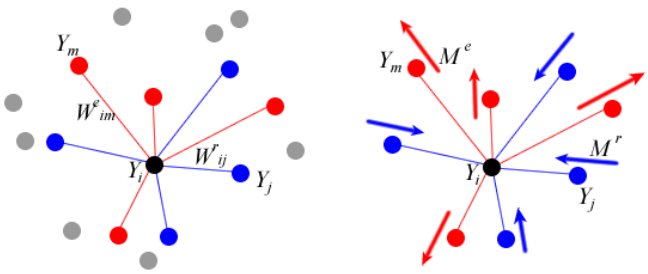

Fig. 1 Left: Data point can be embedded in low-dimensional space using the weights from the intra-class (blue) and inter-class (red) neighborhoods. Right: Solving the constrained optimization problem leads to minimization of the intra-class weights and maximization of inter-class weights. Theoretically, weight values are proportional to the inverse covariance between points.

Motivated by Fisher's discriminative criterion (similar to objective function of LDA [1]), the objective function can incorporate discriminative property that enables the compacting of intra-class neighborhood points and the repulsion of inter-class neighborhood points. The intra-class cost function in Eq. (9) can be minimized so that the overall

weighted pairwise distances between intra-class neighbors in embedded space are reduced. In the spirit of Fisher's criterion, the inter-class cost function should be maximized (see Fig. 1). Since the total sum of weights in subjected to $\sum_j W_{ij} = 1$ or $tr(\mathbf{W}) = \mathbf{I}$, the inter-class cost function in Eq. (10) can be formulated as a *local constraint*,

$$\mathbf{A}^T \mathbf{X} \mathbf{M}^e \mathbf{X}^T \mathbf{A} = \mathbf{I} \tag{11}$$

In order to maintain global rotational invariance of the embedding structure, another constraint is used to further optimize the objective function. From Step 3 of the LLE algorithm, rotational invariance is achieved by subjecting $\sum_{i=1}^{N} \mathbf{y}_i \mathbf{y}_i^T / N = \mathbf{I}$ or $\mathbf{Y}\mathbf{Y}^T / N = \mathbf{I}$. Thus, the *global constraint* is formulated as

$$\mathbf{A}^T \mathbf{X} \mathbf{X}^T \mathbf{A} / N = \mathbf{I} \tag{12}$$

Modelling a constrained optimization problem involving two constraints will result in a Lagrangian with two Lagrange multipliers. Thus, two separate sub-problems are formulated:

$$L_\ell(\lambda_\ell, \mathbf{A}) = \mathbf{A}^T \mathbf{X} \mathbf{M}^r \mathbf{X}^T \mathbf{A} + \lambda_\ell (\mathbf{I} - \mathbf{A}^T \mathbf{X} \mathbf{M}^e \mathbf{X}^T \mathbf{A}) \tag{13}$$

$$L_g(\lambda_g, \mathbf{A}) = \mathbf{A}^T \mathbf{X} \mathbf{M}^r \mathbf{X}^T \mathbf{A} + \lambda_g (N\mathbf{I} - \mathbf{A}^T \mathbf{X} \mathbf{X}^T \mathbf{A}) \tag{14}$$

By forming two equations from their derivatives and solving them simultaneously, it can be easily shown that $\lambda_\ell = \lambda_g = \lambda$ based on the constraints in Eq. (11) and Eq. (12). Thus, the corresponding unified Lagrangian is

$$L(\lambda, \mathbf{A}) = \mathbf{A}^T \mathbf{S} \mathbf{A} + \lambda((N+1)\mathbf{I} - \mathbf{A}^T \mathbf{C}_\ell \mathbf{A} - \mathbf{A}^T \mathbf{C}_g \mathbf{A}) \tag{15}$$

where

$$\mathbf{S} = \mathbf{X} \mathbf{M}^r \mathbf{X}^T \tag{16}$$

$$\mathbf{C}_\ell = \mathbf{X} \mathbf{M}^e \mathbf{X}^T \tag{17}$$

$$\mathbf{C}_g = \mathbf{X} \mathbf{X}^T \tag{18}$$

By setting its gradient to zero,

$$\frac{\partial L}{\partial \mathbf{A}} = 0 \Rightarrow 2\mathbf{S}\mathbf{A} - \lambda\left[2\mathbf{C}_\ell \mathbf{A} + 2\mathbf{C}_g \mathbf{A}\right]$$

and rewriting it as a generalized eigenvalue problem:

$$\mathbf{S}\mathbf{A} = \lambda(\mathbf{C}_\ell + \mathbf{C}_g)\mathbf{A} \tag{19}$$

Note that matrices $\mathbf{S}$, $\mathbf{C}_\ell$ and $\mathbf{C}_g$ are all symmetric and semi-positive definite. The optimal embedding $\mathbf{A}$ is solved by taking d eigenvectors associated with the d smallest eigenvalues ($\lambda_1 < ... < \lambda_d$), where $d < D$.

If $(\mathbf{C}_\ell + \mathbf{C}_g)$ is invertible, Eq. (19) can be easily simplified to a standard eigenvalue problem where optimal columns of $\mathbf{A}$ are the eigenvectors of the matrix $(\mathbf{C}_\ell + \mathbf{C}_g)^{-1}\mathbf{S}$.

In a more generalized case, a constraint tuning parameter $\beta = \{\beta \mid 0 \le \beta \le 1\}$ can be introduced to allow both local and global constraints to be adjusted according to importance:

$$\mathbf{S}\mathbf{A} = \lambda(\beta\, \mathbf{C}_\ell + (1-\beta)\, \mathbf{C}_g)\mathbf{A} \tag{20}$$

For equal contributions between both constraints, we can fix $\beta = 0.5$.

III. Face Recognition in Video

In this paper, face recognition is performed entirely on video sequences, i.e. both training and test data consist of video sequences. In order to realize *video-to-video* recognition, one possible simplified configuration is to reduce it to a *still-to-video* recognition – where still video frames from the training samples are used to recognize test videos. Here, the methods used for extraction of video frame training set and classification of test video sequences are briefly described.

A. Extraction of Video Exemplar Set

The abundance of images in video poses a methodological challenge. While *still-to-still* face recognition is a straightforward matching of a test image to a gallery of training images, it is an ill-posed problem for video sequences.

Similar to approaches [7][9], the subject in each training video is represented by a set of face images extracted from the video, called an *exemplar set*.

For general notation, let a training face sequence $X = \{x_i^\omega \mid i = 1,...,N\}$ be represented as a set of N image frames belonging to one of c subject classes $\{\omega \mid \omega \in \{1,...,c\}\}$. Its associated extracted exemplar set is denoted as

$$E_\omega = \{e_j^\omega \mid j = 1,...,M; e_j^\omega \in X\} \tag{21}$$

for the ω-th class, where the number of exemplars extracted from each video, $M < N$.

Face images are projected to LLE space (see Section II-A) in order to learn a low-dimensional embedding from their original high-dimensional data space. Then, extraction of exemplars is performed by Hierarchical Agglomerative Clustering (HAC) [19], a hierarchical method of partitioning data points by a nested set of partitions represented by a cluster tree, or dendrogram. The HAC procedure starts with N singleton clusters and terminates when all points belong to a single large cluster. Then, the required number of clusters, M is selected by partitioning at the appropriate level of the constructed dendrogram. Finally, the nearest cluster centers are chosen to form the exemplar set.

The exemplar training set forms the basis for learning the optimal embedding by the proposed NDMP algorithm.

B. Classification of Video Sequences

Most conventional still-image face recognition systems evaluate performance by accuracy or error rates, which can simply be computed based on the number of correctly or incorrectly identified test images in a particular test set. In video-based evaluation, this can be extended to a majority voting scheme, where the face in every video frame is identified, and then a majority vote is performed to decide on the identity of the person in the sequence.

In this work, we propose evaluation using two probabilistic classification schemes, which are able to produce more reliable measures than the majority vote method.

1) Probabilistic majority voting scheme: Recognition probability scores are combined across frames to take a vote in each frame. The class with the majority total vote

is classified as the subject identity of the test video sequence. The exemplar probability scores are computed using the reciprocal of a *L2*-norm distance measure between a test video frame, x and a training exemplar, e, followed by normalization by the total sum of probability scores,

$$\hat{P}(x,e) = \frac{1/\|d(x,e)\|}{\sum_{\omega=1}^{c}\sum_{j=1}^{M} 1/\|d(x,e_j^{\omega})\|} \tag{22}$$

By means of classifier combination, the class probability score is easily obtained by averaging all j-th exemplar probability scores belonging to the same class.

2) Bayes maximum a posteriori (MAP) classifier: This evaluation method determines the subject identity of a probe video X by the MAP solution of a Bayes classifier,

$$\begin{aligned}\omega^* &= \arg\max_{\omega} P(\omega \mid x_{1\ldots N}) \\ &= \arg\max_{\omega} \frac{P(\omega)P(x_{1\ldots N} \mid \omega)}{\sum_{\omega'}^{c} P(x_{1\ldots N} \mid \omega')P(\omega')}\end{aligned} \tag{23}$$

where $P(\omega \mid x_{1\ldots N})$ is the posterior probability of the constructed Bayes classifier. $P(x_{1\ldots N} \mid \omega)$ and $P(\omega)$ are the class likelihood calculated based on Eq. (22) and the class prior, respectively.

IV. Experimental Results and Analysis

The performance of the proposed NDMP manifold learning approach was evaluated in a video-based face recognition setting using the Honda/UCSD Video Database [13].

Fig. 2 Sample extracted face images from the Honda/UCSD Video Database

The Honda/UCSD database is one of the most commonly-used benchmark data set for face recognition in video. We use its first data set, which consists of 59 video sequences of 20 different persons, divided into a training subset (of 20 persons) and a test subset (of 19 persons). Each video contains about 300-500 image frames of large pose and scale variations, significant 2-D (in-plane) and 3-D (out-of-plane) head rotations, and also expression variations. Sample extracted face images of a same person are shown in Fig. 2.

Image training and test sets were constructed by detecting faces in each video sequence using the Viola-Jones face detector [18] and the Incremental Visual Tracker (IVT) [14], which is more robust towards out-of-plane head movements. The detected faces were resized to 32×32 gray-scale images. Following typical protocol, the detected face images are histogram-equalized to even out illumination effects, and no further pre-processing is applied. For each subject, one video is used for training, and the remaining videos are for testing.

To ensure an extensive evaluation is conducted on a single large data set, Q video sequences of length L each for every subject were randomly sampled from the test subset. This allows us to generate $c*Q$ number of randomly-selected video sequences for evaluation.

TABLE I
Performance Comparisons on the Honda/UCSD Database Between Various Projection Methods

Method	Recognition Rate (%)	
	Probabilistic Voting Scheme	MAP Classifier
PCA	48.95	64.93
LDA	54.21	69.64
LPP	35.26	55.21
OLPP	42.63	61.59
MFA	15.79	47.49
NPE	46.05	72.10
$\text{NDMP}_{\beta=0.5}$	68.95	83.72

The proposed NDMP method is compared against other classical approaches (PCA, LDA) and recent state-of-art approaches (LPP, OLPP, MFA, and NPE). The recognition rates based on probabilistic voting scheme and Bayes classifier are summarized in Table 1. The optimal number of feature dimensions for all tested methods were determined empirically. As observed, the NDMP method performed the best under both classification schemes.

For all experiments, we use $Q = 20$ and $c = 19$ (from test subset), generating (19*20) 380 randomly-sampled video sequences, each containing $L = 200$ frames. We fix the number of exemplars used per class as $M = 7$ while the number of intra-class, k and inter-class neighbors k' is 6. Fig. 3 shows some sample face exemplars extracted from the training video sequences.

Fig. 3 Sample face exemplars extracted from training video sequences

To evaluate the performance of the NDMP algorithm in a rank-based identification setting, we present the recognition rates using a cumulative match curve (CMC), as shown in Fig. 4. Classification is performed with probabilistic majority voting scheme. For all rank-n top matches for n={1,...,8}, features extracted by the NDMP algorithm yielded higher recognition rates. The CMC reinforces the superiority of the NDMP algorithm compared to other state-of-art methods.

Classical approaches like PCA and LDA did not fare well in our experiments due to their inability to learn nonlinear manifolds of large face variations. Interestingly,

discriminative methods such as NDMP and LDA tend to improve their recognition rates remarkably with increasing rank. Using the MAP classifier which considers temporal dependency between video frames, it can be observed that the neighborhood preserving property based of the NDMP and NPE methods produced better results than the other methods.

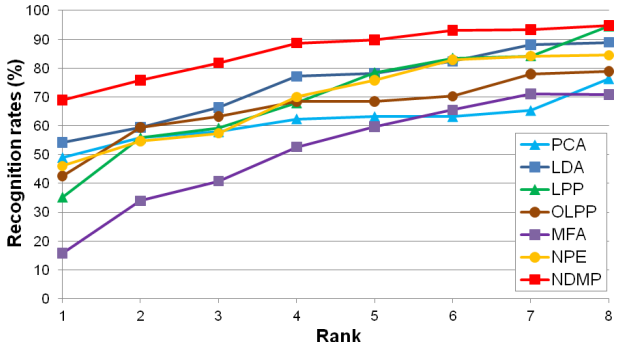

Fig. 4 Cumulative match curve of various manifold learning methods based on classification by probabilistic majority voting.

The constraint tuning parameter, β in our generalized version of NDMP can be tuned to different values between 0 and 1 to balance between the contribution of the local neighborhood constraint and global invariance constraint. Table II compares the recognition performance using the NDMP algorithm with different values of β.

TABLE III
PERFORMANCE OF NDMP WITH DIFFERENT β VALUES

Method (and β value)	Recognition Rate (%)	
	Probabilistic Voting Scheme	MAP Classifier
$NDMP_{\beta=0.0}$	65.26	79.73
$NDMP_{\beta=0.25}$	66.58	80.92
$NDMP_{\beta=0.5}$	68.95	83.72
$NDMP_{\beta=0.75}$	68.42	83.05
$NDMP_{\beta=1.0}$	67.63	82.19

From this, we can observe that both local and global constraints are equally important in projection methods. Tuning to either side of the value range will result in either the lack of invariance in global embedding structure, or the lack of neighborhood discriminative power. However, it remains inconclusive if an optimal value for β can be determined.

V. CONCLUSION

We have presented a novel supervised manifold learning algorithm called Neighborhood Discriminative Manifold Projection (NDMP), for face recognition in video. The NDMP constructs a discriminative eigenspace projection of a high-dimensional face manifold based on both local neighborhood geometry and global invariance structure. The local geometry is preserved through the use of intra-class and inter-class neighborhood information, while global structure is preserved by imposing rotational invariance. Empirical results on a large video data set demonstrated the superiority of the proposed approach compared to classical and recent state-of-art methods. NDMP is also more robust in handling large face variations inherent in video sequences.

In future work, the NDMP method can be generalized from the linear form to a nonlinear form in kernel feature space. For better practicality, our approach should be further tested with larger face databases or real-life video data in order to further substantiate the robustness and capability of our approach. Also, the effects of video sequence length and video frame resolution on the recognition rates can be further investigated.

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